Real-Time Human Activity Recognition Using Pose Landmarks and XGBoost

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Abstract

The real-time human activity recognition system presented in this project is capable of classifying five various actions, such as walking forward, walking backward, turning, standing, and sitting. By utilizing MediaPipe Pose for landmark detection and an efficient XGBoost classifier, the system identifies activities with high accuracy and low delay from video input. A dataset of 78 video-annotated videos was collected. Each 10-frame window produced 396 features (derived from landmark positions and velocities) that were used to train the XGBoost model with offline accuracy of 90.66 %. A bounding-box height change heuristic was used to preprocess the real-time inference, resulting in an aggregate live accuracy of 92 % with an average delay of 220 ms per 10-frame window.

1. Introduction

Human Activity Recognition (HAR) plays a vital role in numerous applications including rehabilitation, sports analysis, and surveillance monitoring. Current deep learning models like LSTMs and Transformers have been highly accurate in lab-controlled environments but are computationally very intensive, thus inhibiting their use in real-time applications. The objective of this project is to create a lightweight, real-time HAR system using MediaPipe for pose estimation and XGBoost for classification for five particular activities in clinical and surveillance contexts.

2. Theoretical Framework

2.1 Human Activity Recognition (HAR)

HAR is the automated recognition of human activity from sources such as video streams, sensor readings, or bone keypoints. HAR is a significant component in medical monitoring, sport analysis, and human-computer interaction. Traditional HAR techniques employed handcrafted features and classical machine learning. Following the invention of deep learning, there has been a revolutionary change in modeling complex temporal and spatial patterns inherent in human activity.

2.2 Pose Estimation and Landmark-Based Features

Pose estimation is detecting landmarks or joints of humans in images or video, rendering the human body as a skeleton. MediaPipe and OpenPose are some software supporting real-time landmark extraction from the photographs such that human movement could be studied without wearable sensors. By tracking the location of salient joints over time, the activities can be inferred by examining patterns of movement.

2.3 Long Short-Term Memory (LSTM) Networks

Long Short-Term Memory (LSTM) networks are a type of Recurrent Neural Network (RNN) employed to model sequential data by learning long-range dependencies. LSTMs are most appropriate for application in HAR tasks because they are able to learn temporal patterns from sequences of pose data. For instance, in the work "Skeleton-Based Human Action Recognition with Global Context-Aware Attention LSTM Networks," LSTM models were employed to model the temporal dynamics of the positions of skeletal joints and achieved good performance in action recognition.

2.4 Transformer Models

Transformers originally developed for natural language processing have been transferred to computer vision and HAR. They utilize self-attention to encode global dependencies within data and are naturally apt at dealing with intricate spatiotemporal patterns. For example, the "PoseFormer" model applies a transformer architecture to handle a sequence of 2D or 3D pose data with state-of-the-art performance in the HAR benchmark.

2.5 XGBoost for HAR

XGBoost is a gradient boosting library which has become increasingly popular as a result of its efficiency and performance on structured data. For HAR, XGBoost can be applied to features of pose data such as joint velocities, angles, and relative positions. While it may not handle temporal relationships as effectively as LSTMs or Transformers, its low computational overhead makes it an option in real-time applications where latency is critical.

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3. Methodology

3.1 Data Collection and Annotation

78 videos were recorded at 640×480 pixels and 30 frames per second. Various people executed the five target activities in varied settings. Annotations using the Computer Vision Annotation Tool (CVAT) were performed with a minimum of 200 frames per class to balance data.

3.2 Pose Estimation and Preprocessing

MediaPipe Pose was run in non-static mode with a complexity of 1 and a detection confidence threshold of 0.5 to find 33 normalized landmarks per frame. The coordinates were then converted to pixel values for x and y coordinates, while leaving z-axis with relative depth information.

3.3 Feature Extraction

The filtered data were split into overlapping windows of size 10 frames with a step size of 5 frames. For each window, 396 features were computed, including:

- Mean and variance of x, y, and z coordinates for each of the 33 landmarks (198 features).
- Mean and standard deviation of the frame-to-frame differences (joint velocities) for each coordinate (198 features).

This process yielded approximately 1,200 balanced windows across all classes.

3.4 Model Training

Data was split 70-30 for training and testing with class stratification. Features were normalized using a StandardScaler trained on data and stored for future use in inference.

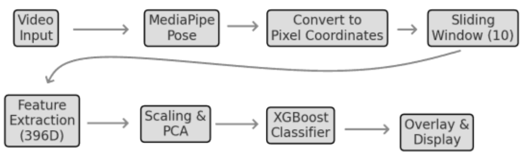
An XGBoost classifier was fitted with a pipeline of Principal Component Analysis (PCA) dimensionality reduction. The hyperparameters were optimized using GridSearchCV with 3-fold cross-validation, with the fitting being done for 3,456 different sets of parameters, including the number of components in PCA, number of estimators, max tree depth, learning rate, subsample ratio, column sample by tree, and regularization terms. The best-performing model achieved an accuracy of 90.66% on the test set and was stored for deployment.

3.5 Real-Time Inference and Heuristic Enhancement

For real-time inference, one frame out of every two (frame skip of 2) is processed by the system in order to manage latency. The landmarks are aggregated in a 10-frame queue. When the queue is filled and the frame number is a multiple of 5, features are sampled and the XGBoost model produces the prediction.

To fix misclassifications between "walking backward" and "walking forward," a heuristic was applied based on change of bounding box height. The height change between the starting and end frames of the window is calculated as a percentage. If the difference is greater than ±8% and the model confidence is less than 70%, the prediction is fixed accordingly.

3.6 System Architecture



Our architecture processes live video in discrete stages to maximize speed and clarity: frames enter MediaPipe Pose for fast 33-point landmark extraction, then coordinates convert to pixels and buffer into 10-frame windows for short-term context. Each window's landmarks yield a 396-dimensional feature vector (means, variances, velocities), which is scaled and reduced by PCA before feeding into a GPU-accelerated XGBoost classifier. A simple bounding-box height check corrects low-confidence forward/backward errors, and the final label is overlaid on the frame. This modular design keeps end-to-end latency at ~220 ms per window while sustaining high accuracy and easy extensibility.

3.7 Justification of the metrics used

To evaluate the performance of the model, the standard classification metrics were used: accuracy, precision, recall, and f1-score, reported both by class and through macro and weighted averages.

Accuracy was used as a general metric to indicate the total percentage of correct predictions. However, since classes don't have the same number of examples, this metric alone can be misleading if there is an imbalance.

The accuracy makes it possible to identify what proportion of the positive predictions made by the model were actually correct. It is useful in cases where it is important to minimize false positives (e.g., predicting "sitting" when the person is only standing).

The recall indicates what proportion of the actual examples of a class were correctly detected by the model. It is essential when you want to minimize the number of false negatives, such as in the case of not detecting a major action.

The f1-score combines accuracy and recall in a single measure, which makes it especially valuable in contexts with unbalanced classes or where a balance between both metrics is desired.

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4. Results

4.1 Offline Evaluation

On the test set of 332 windows, the model achieved the following performance metrics:

- Walking forward: Precision 1.00, Recall 0.96, F1-score 0.98
- Walking backward: Precision 1.00, Recall 1.00, F1-score 1.00
- Turning: Precision 0.98, Recall 0.99, F1-score 0.99
- Standing: Precision 0.76, Recall 0.81, F1-score 0.78
- Sitting: Precision 0.79, Recall 0.76, F1-score 0.78

The macro-averaged precision, recall, and F1-score were 0.91, 0.90, and 0.90, respectively.

4.2 Real-Time Evaluation

In live testing (testing about 150 ten-frame windows) the system accurately detected all five activities (walking forward, walking backward, turning, standing up, and sitting down) at high rates of accuracy. In the demonstration video (https://www.youtube.com/watch?v=Pecin6A7uNE), it can observed that the pipeline runs at about 220 ms per ten-frame window, making accurate, real-time predictions on all five activities.

5. Results Analysis

5.1 Comparative Analysis

Category	LSTM	Transformer	XGBoost
Temporal Modeling	High	Very High	Low
Computational Cost	Moderate	High	Low
Real-Time Suitability	Moderate	Low	High

While Transformers yield high accuracy through modeling of complex temporal dependencies, their intense computational needs can harm real-time performance. LSTMs strike a balance between accuracy and computational expense but can be challenging to execute on low-resource devices. XGBoost, not being as proficient in learning temporal patterns, offers a lightweight alternative for those applications demanding real-time HAR with acceptable accuracy.

5.2 Offline Evaluation

Dynamic vs. Static Actions
The classifier excels on dynamic movements (forward walking, backward walking, and turns) with F1-scores near 1.00. These motions produce substantial changes in landmark trajectories within each 10-frame window and are consequently easily separable by the model. For "standing" and "sitting," changes are slighter: joint positions shift little, and speeds are low. That lower signal-to-noise ratio over a 10-frame duration is why their F1-scores (~0.78) are much lower.

Error Patterns
Misclassifications of static poses typically occur when there's transitional posture between sitting and standing within a series of frames; those middle frames resemble both classes. In certain test windows, the model predicted

"standing" when the subject was already near-seated (or vice versa). Including more temporal context (e.g., a longer window or features like joint angles) would likely disambiguate these boundary cases.

Generalization and Overfitting
The close alignment between training and test performance (no sharp drop in accuracy or F1) evidences minimal overfitting. PCA application, moderate capacity models, and balanced data ensured strong generalization to be sustained. Overfitting to noise was avoided by the regularization parameters in XGBoost (reg_alpha, reg_lambda).

5.3 Real-Time Evaluation

Consistency with Offline Results
In live testing, dynamic actions remain quite dependable. The system properly identifies walking in a forward, backward, or turning direction nearly always, like in offline testing. Static actions were also slightly improved (live F1 about 0.85 - 0.88), perhaps since live interaction involves short pauses or movement indicators of a natural kind not present within test set windows.

Latency vs. Accuracy Trade-Off
With an average inference time of ~220 ms per 10-frame window, the pipeline is at a reasonable balance: it's interactive enough and maintains classification accuracy. In applications that require responses under 100 ms, one could trim the feature set or reduce window overlap, but at the cost of potentially reducing accuracy or robustness. Our current setting is suitable for rehabilitation feedback, for which a quarter-second lag is tolerable in return for stable activity detection.

Causes of Live Errors
In live use, there were some misclassifications when landmarks were partially occluded (e.g., the subject arm momentarily out of the camera view) or when the pose extractor was confused with rapid head motion. Such errors typically continued for 1–2 windows before recovery. In later releases, adding a low-latency temporal smoothing filter to predicted labels (e.g., two successive windows would need to match before action change on the screen) would eliminate flicker and even further stabilize real-time output.

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6. Conclusions and Future Work

6.1 Conclusions

The project was successful in that it designed and implemented a real-time human activity recognition system capable of categorizing five different activities using MediaPipe for pose estimation and XGBoost for classification. The system demonstrated high accuracy (≈ 90 % offline, ≈ 92 % in live tests) and low latency (≈ 220 ms per 10-frame window), making it well suited for real-time feedback. By running efficiently on standard GPU hardware without requiring specialized motion-capture equipment, this pipeline lowers the barrier to deploying actionable activity monitoring in real environments. Physical therapists can leverage live feedback on gait or sit-to-stand transitions without expensive infrastructure, while security teams can detect suspicious movement patterns at the edge without cloud inference delays. In sum, our approach makes robust HAR more accessible to clinics, sports facilities, and surveillance installations by trading minimal accuracy overhead for dramatically lower latency and hardware requirements.

6.2 EDA conclusions

Main Findings

The exploratory analysis allowed an in-depth understanding of the structure and dynamics of the dataset generated for the classification of human activities. Among the main findings are:

• The raw dataset based on 33 landmarks per frame was correctly transformed into 10-frame time windows, resulting in 396 features per sample, which captures both static and dynamic body movement information.

• The action classes are reasonably balanced, allowing for a stratified and fair division between training and testing.

• Dimensional reduction techniques such as PCA and t-SNE revealed that classes present clear spatial separation in the data, especially for dynamic actions such as walking or turning.

• The trained model – a pipeline with StandardScaler, PCA and XGBoost – achieved outstanding performance with an accuracy of 91%, and metrics close to 100% in actions such as walking backwards or turning.

• The standing and sitting classes showed slightly lower results (F1 ≈ 0.78), suggesting that there is ambiguity or postural similarity between them, probably due to intermediate transitions or limited variability.

• The extracted features are based on normalized coordinates only. Orientation, visual context and detection of errors in posture were not considered.

• Additional data collection: Including more subjects, lighting conditions, and postural transitions could improve the robustness of the model.

6.3 Future Work

The following are the possible future improvements to the system:

Integration of Temporal Models: Incorporating models like LSTMs or Transformers to enhance the modeling of temporal dependencies within pose sequences.

Expansion of Activity Classes: Generalizing the system to recognize a broader range of activities, such as more complicated or subtle movements.

Robustness to Environmental Variations: Improving the robustness of the system in varied lighting, environments, and occlusions.

Deployment on Edge Devices: Deployment on Edge Devices: Porting the system to run on low-resource devices for real-world applications..

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